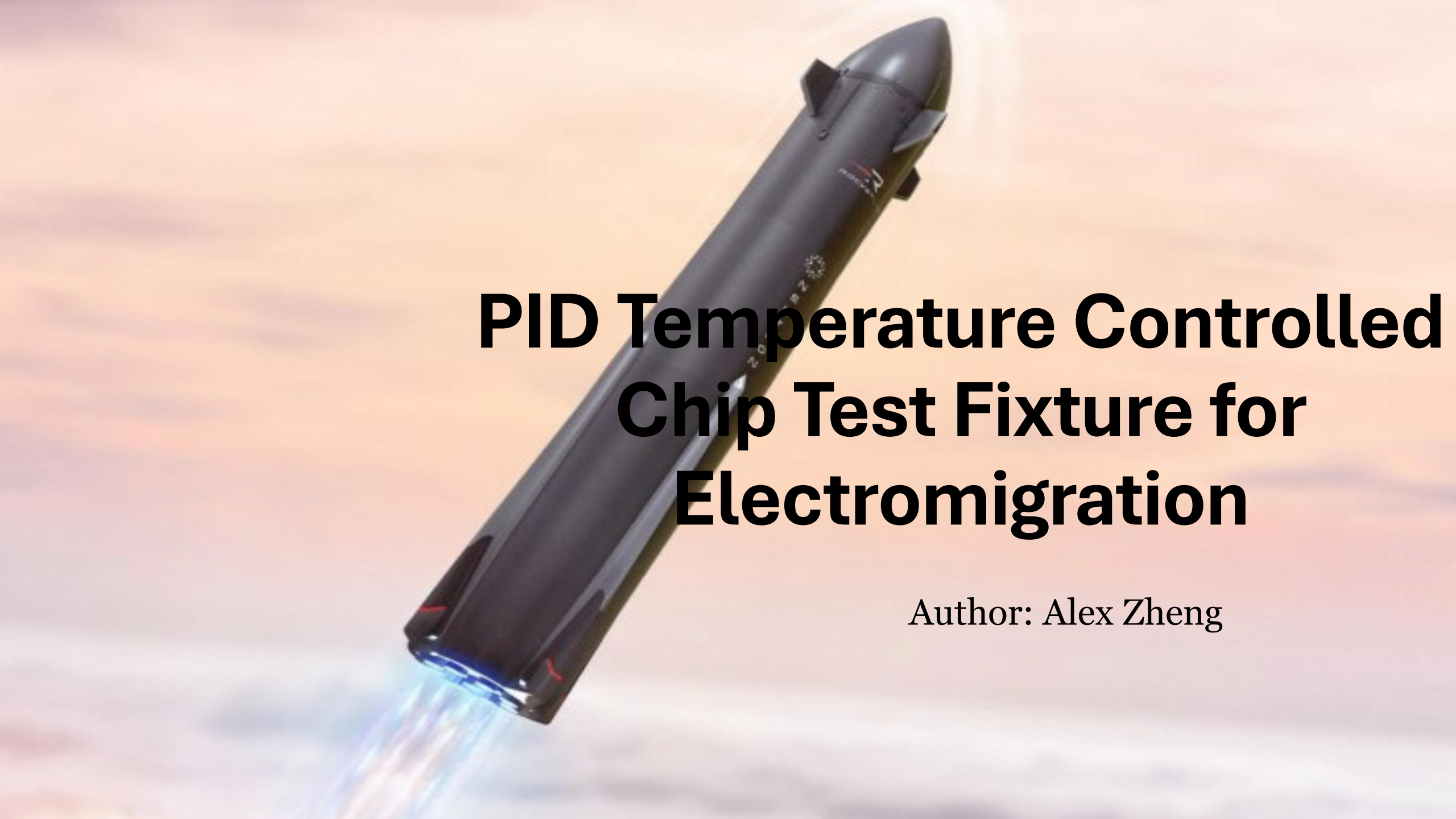




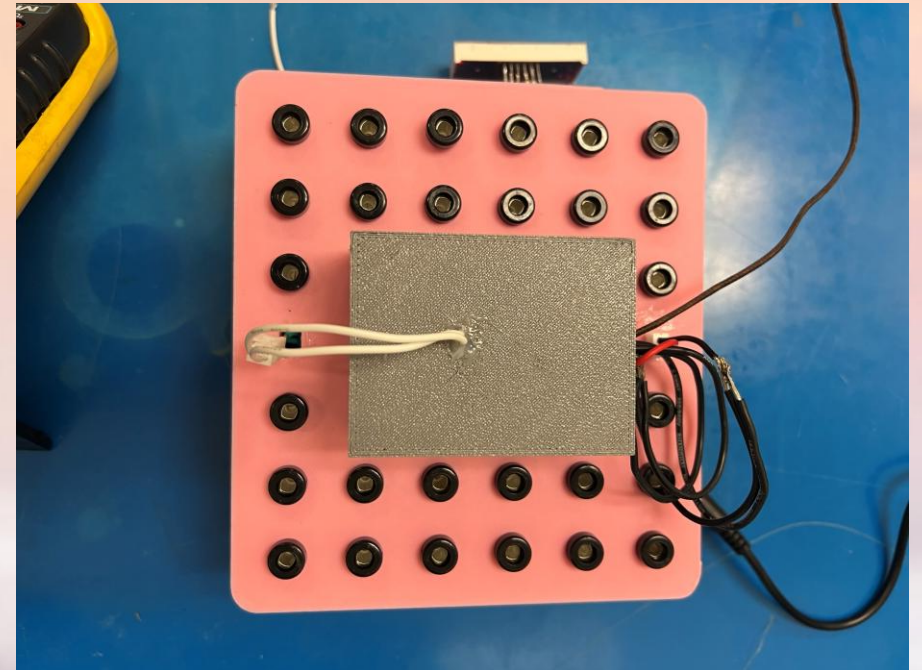
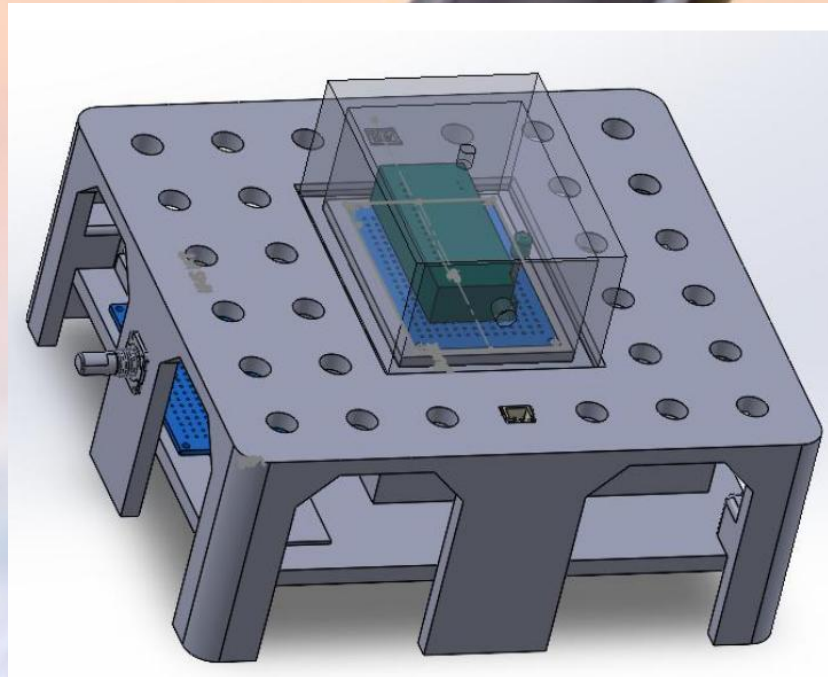
PID Temperature Controlled Chip Test Fixture for Electromigration

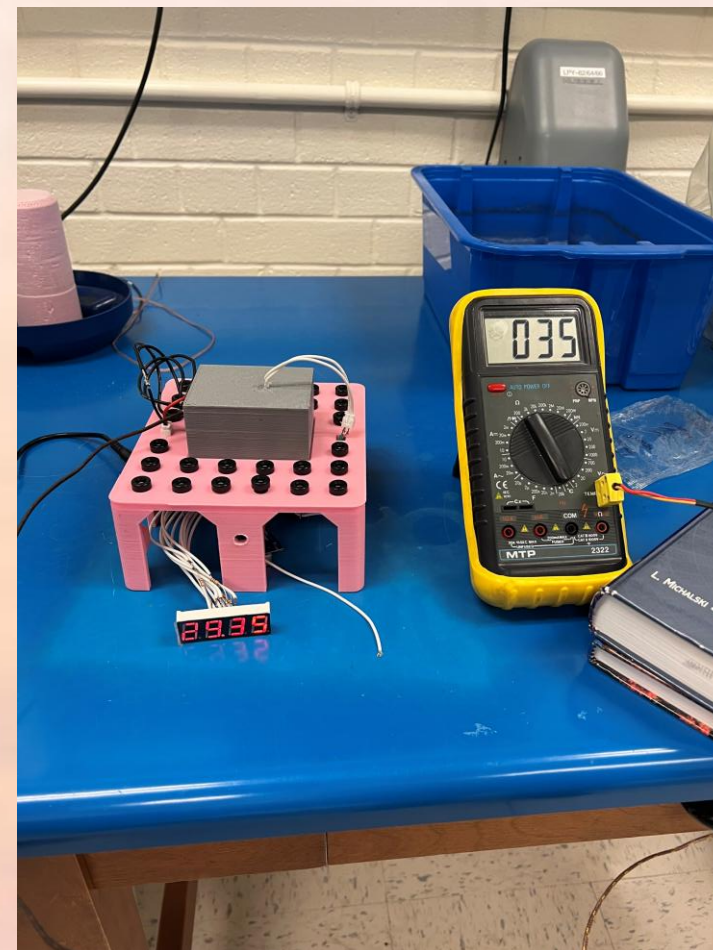
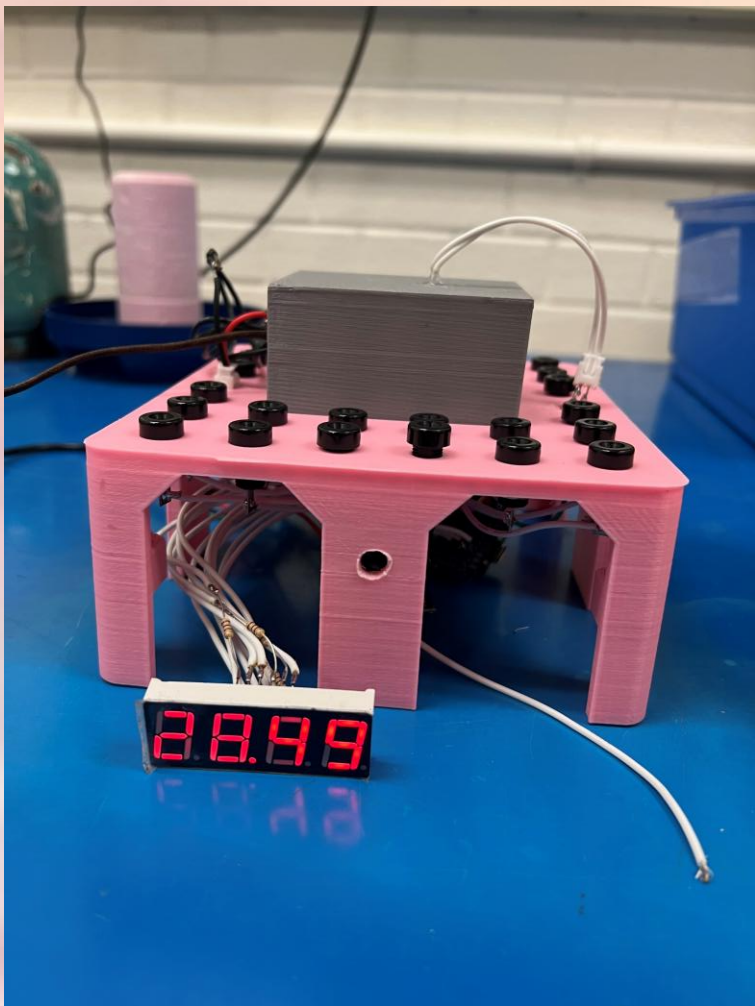
Author: Alex Zheng

PID-Temperature Controlled Test Bench for ICs

- Electromigration Laboratory
 - Ultrasonically soldered gold wire bonds to test
- Temperature $\propto$ Resistance ($t = 40^\circ\text{C}$)
- Black's equation for semiconductor circuits due to electromigration:

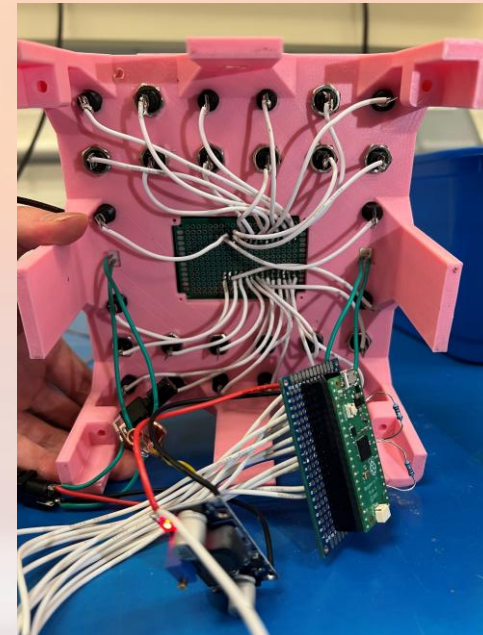
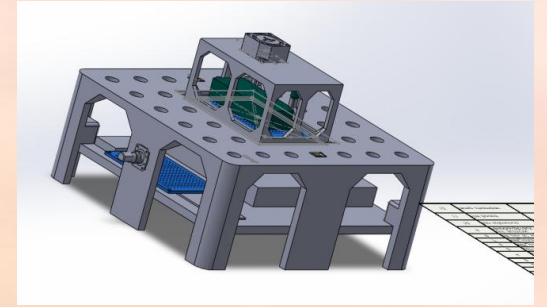
$$\text{MTTF} = \frac{A}{j^n} e^{\left(\frac{Q}{kT}\right)}$$





System Constraints/Technical Challenges

- Design speed
- Size and Price:
 - BoM cost in research lab, increased 3D printing costs, testing costs for more surface area
- Modularity
 - Plug and play
 - Single unit
- Hardware integration into module
 - Mechanical
 - Electronics
 - Control
- Predetermined chip carrier
 - Accommodates off the shelf component



Mechanical Design

- Lengthiest process
 - System Modelling in CAD
 - Test bench and component design/size optimization
 - Tolerance and edge cases
 - Manufacturing (3D printing tests, prototypes, and final)
 - Friction fit tests
 - Hardware Inference
 - Reliability tests

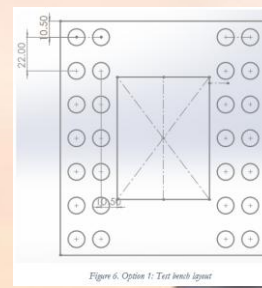


Figure 6: Option 1: Test bench layout

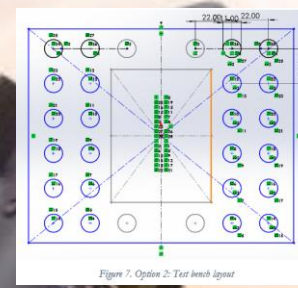


Figure 7: Option 2: Test bench layout

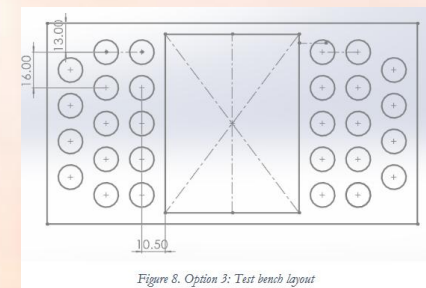


Figure 8: Option 3: Test bench layout

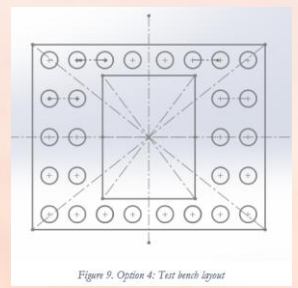
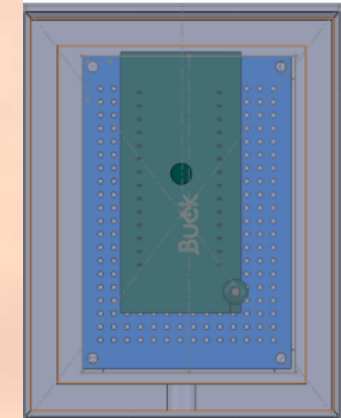
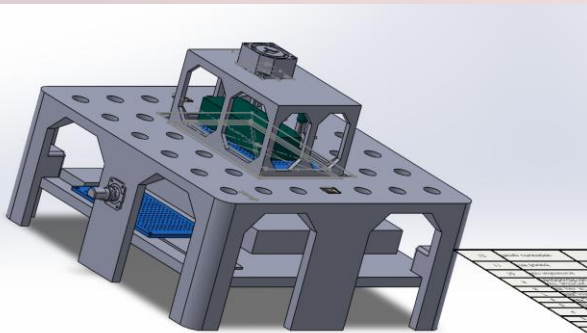


Figure 9: Option 4: Test bench layout



- Acrylic Glass
- concrete
- cork
- cotton
- plastic insulated foam
- Epoxy
- PLA → PETG

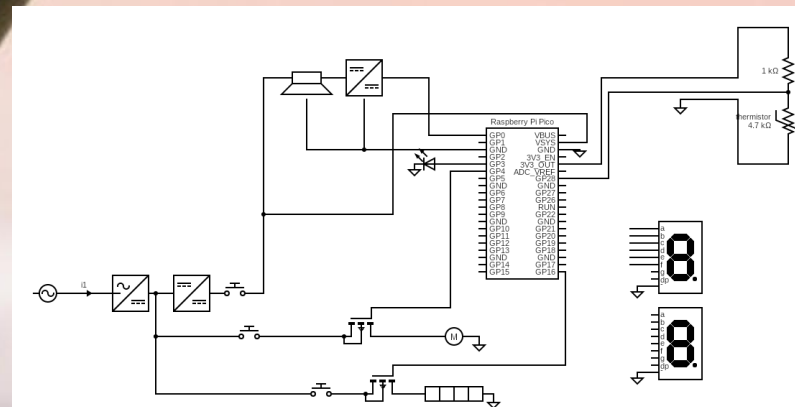
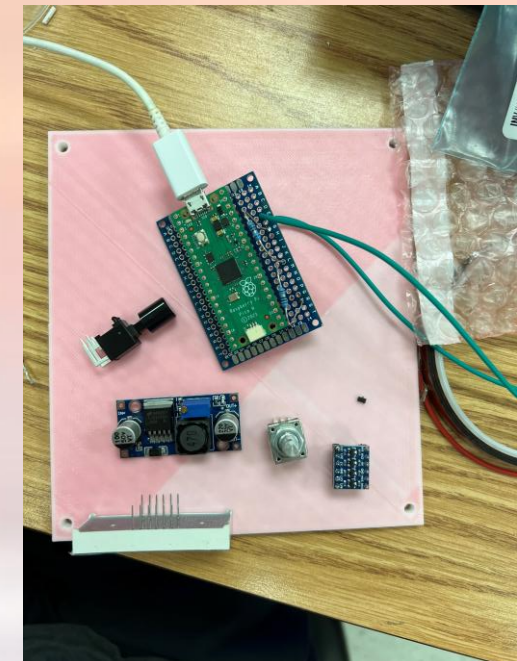
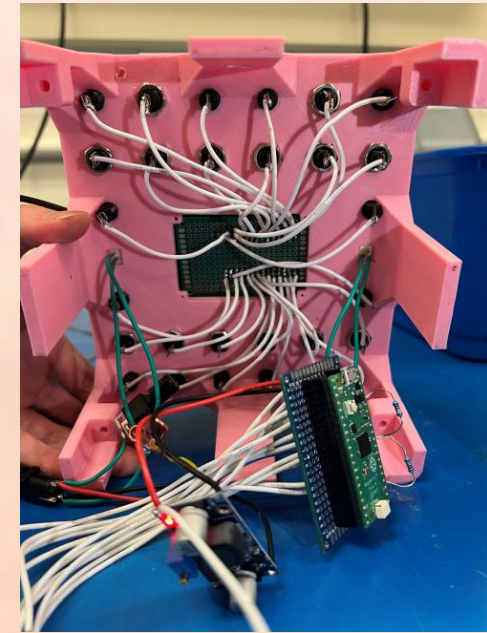


Electronics

- Idea-to-component Process
- What sensor? -> Thermistor
- Power conversion
 - Fan, heater, microcontroller
- BoM generation
 - Off the shelf electronics
 - How is a part actually used?
- Soldering
 - Mechanical integration

Fan: 12V, 0.05A $I \leq 0.08A$,
 $0.06W \leq P \leq 0.96W$ from datasheet
 $0.062IV \leq 0.96 \Rightarrow P \leq 1W$
 $0.08 \leq I \leq 0.96$
 $5mA \leq I \leq 80mA$
 Heater: 12V, 3W $P \leq 6W$
 $3 \leq IV \leq 6 \Rightarrow P \leq 6W$
 $3 \leq I \leq 6$
 $0.25A \leq I \leq 0.5A$
 MosFET (PMOS): @ $T_a = 25^\circ C$; $I_{D,max} = 0.3$
 $@ T_a = 25^\circ C$; $I_{D,max} = -0.33A$
 $V_{GS} = 12V$, $V_{DS(4A)} = -2V$
 $V_{GS} = 12V$, $V_{DS} = -12V$
 $V_{GS} = 3.3V$, $12V = -8.7V$ needs C_{GS} & C_{DS}
 P_{Pico} dependant on Code $RA \leq 5W$
 P_{LEDs} , 7-seg, 4-seg, 1-seg, on code $\leq 35W$

$I_{fan} \leq I_{sp}$ $P_{fan} = 0.08 \times 12 = 0.96W$
 $80mA \leq 330mA$ will work
 $I_{heater} \leq I_{sp} \leq I_{heater,max}$
 $P_{heater} \leq 0.33A \times 12V = 3.96W$
 $P_{heater+fan} = 4.92W$
 $P \leq 5 + 3 = 8W$



BOM Level	Part #	Part Name	Description	QTY	Unit	Procurement Type	Cost Pt
	4-BIT COMMON CATHODE	4 bit Red Cathode	0.36inch LED display 7 Segment 1 Bit/2 Bit/3 Bit/4 Bit Digit Tube Red Common Cathode / Anode Digital 0.36 inch Red 7segment	2	Piece	ALIEXPRESS	
	DIFFUSED ANODE	Diffused Anode	50PCS 5mm full color LED RGB red/green/blue Common Cathode/Anode Four feet transparent highlight color light 5mm diode colorful	1	lot(50 PIECES)	ALIEXPRESS	
	LM2596	LM2596a DC-DC step-down	LM2596a DC-DC step-down Power Supply Module 3A Adjustable Step-Down Module LM2596 Voltage Regulator 24V 1.2V 5V 3V	1	Piece	ALIEXPRESS	
	Bi-Directional Level Shifter	Bi-Directional Level Shifter	Spec: 4-Logic Level Shifter Bi-Directional Four-way two-way logic level transformation module for arduino	1	lot(5pcs)	ALIEXPRESS	
	12V 70 DEGREE	12V 70 Degree PTC Ceramic Heater	12V/24V/36V/48V/110V Constant Temperature PTC Ceramic Heating Plate Air Electric Heater Heating Plate 70/110/200/220/230 Degrees	2	Piece	ALIEXPRESS	
	2648-SC0917-ND	SC0917	RASPBERRY PI PICO H R P2040	1	Piece	DIGKEY	
	732-61302011821-ND	61302011821	WR-PHD 2.54 MM SOCKET HEADER 20	2	Piece	DIGKEY	
	2223-CFM-2010CF-0170-292-ND	CFM-2010CF-0170-292	DC AXIAL FAN, 20 MM SQUARE, 10 M 5V	1	Piece	DIGKEY	
	1738-FIT0845-ND	FIT0845	PTC HEATING ELEMENT 5V	2	Piece	DIGKEY	
	PEC11R-4115F-50018-ND	PEC11R-4115F-50018	ROTARY ENCODER MECHANICAL 18PPR	1	Piece	DIGKEY	
	EG5547-ND	P840EEQR1BLK	SWITCH PB DPST 3A 30V	2	Piece	DIGKEY	
	PSAW360-12-82	3989-PSAW360-12-82-ND	AC/DC WALL MOUNT ADAPTER 12V 36W	1	Piece	DIGKEY	
	P19863-ND	0301059-12V-AT-00	FAN 40MM, 3000RPM 12VDC WHITE	1	Piece	DIGKEY	
	CP-046A-ND	P1-046A	CONN PWR JACK 2X5.5MM SOLDER	1	Piece	DIGKEY	
	785-1459-1-ND - Cut Tape (CT)	A03421E	MOSFET P-CH 30V 3A SOT23-3L	2	Piece	DIGKEY	

Software, Calibration, and Controls

- Thermistor (NTC) calibration
 - Steinhart-hart Equation ($T = \{21, 41, 67\}$)
- Microcontroller
 - ADC, Display, rotary encoder, PWM control
- PID (proportion, integration, derivative)
 - PI
 - Tuning:
 - Trial and error
 - Ziegler Nichols
 - Number $\rightarrow$ Percentage
- Code show.

$$T = \frac{1}{\frac{1000 \cdot R}{R_0} + \frac{1}{R_0} \left(\frac{1}{\beta} - \frac{1}{R_0} \right)} \quad [K]$$

```
# Every PIDPeriod seconds the PID calculation is taken
if((time.time() - prevPIDTime) >= PIDPeriod or prevPIDTime == 0):

    # Calculating the duty cycle for our heater using PID control. PID tuning
    currentError = setpoint - Temp

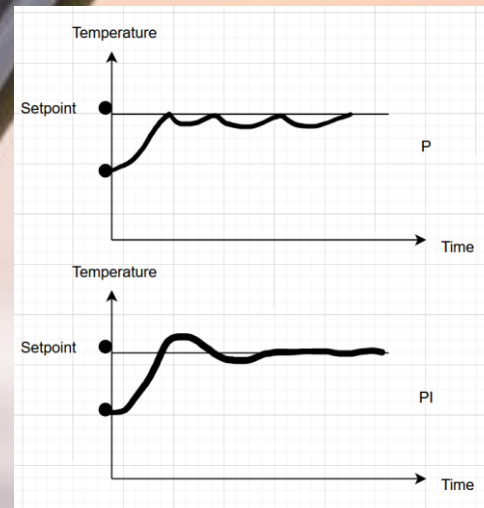
    if(!reachedSetpoint and currentError >= 0):
        reachedSetpoint = true
        integralBound = area

    # use PIDPeriod instead
    # timeChange = (time.time() - prevPIDTime)

    area = area + (currentError + prevError) * PIDPeriod/2
    if(area > integralBound):
        area = integralBound
    #d = (currentError - PrevError) / PIDPeriod

    prevPIDTime = time.time()
    PrevError = currentError

    output = round(Kp*currentError + Ki*area + Kd*d)
    DC = output/calculatedMaxPID
    if(DC > 0.99):
        fet.duty_u16(65535)
    else:
        fet.duty_u16(DC * 65535)
```



Testing and Tuning

- Test plan?
 - Temperature constant > Specific temperature target
- Scope?
 - Graduate researchers refine the system in case-specific manner.
- Achieved prototyping role

